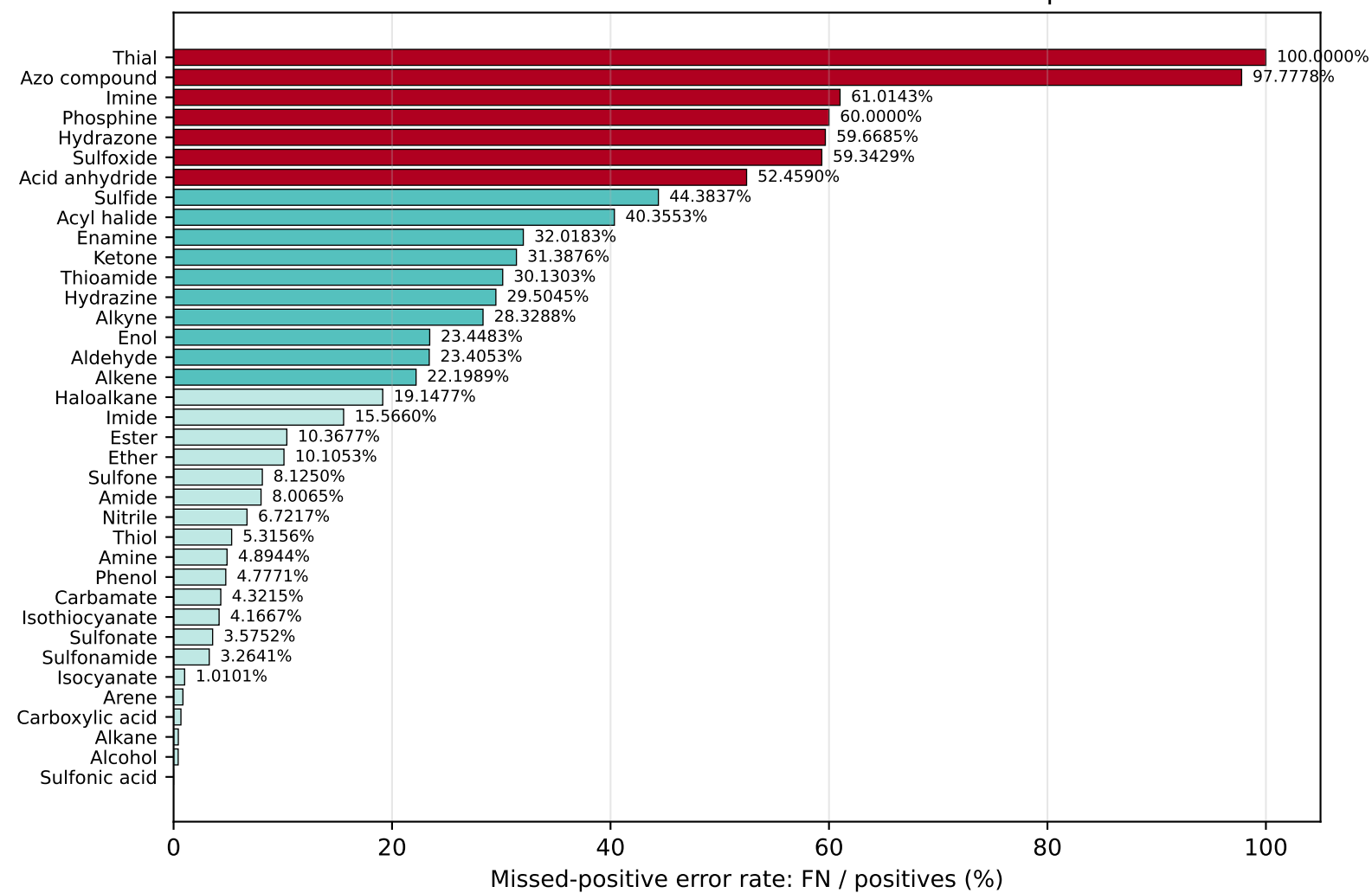
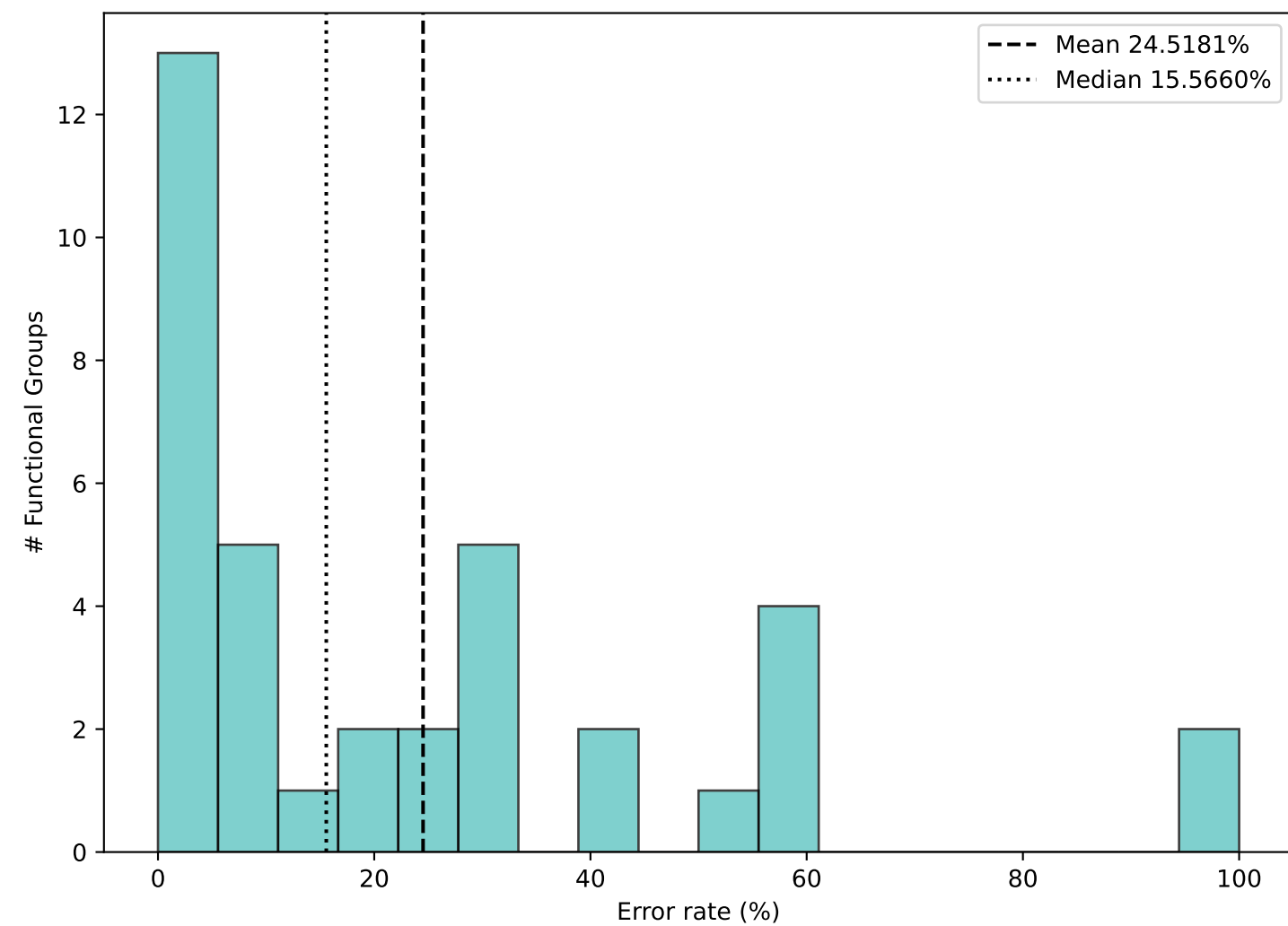


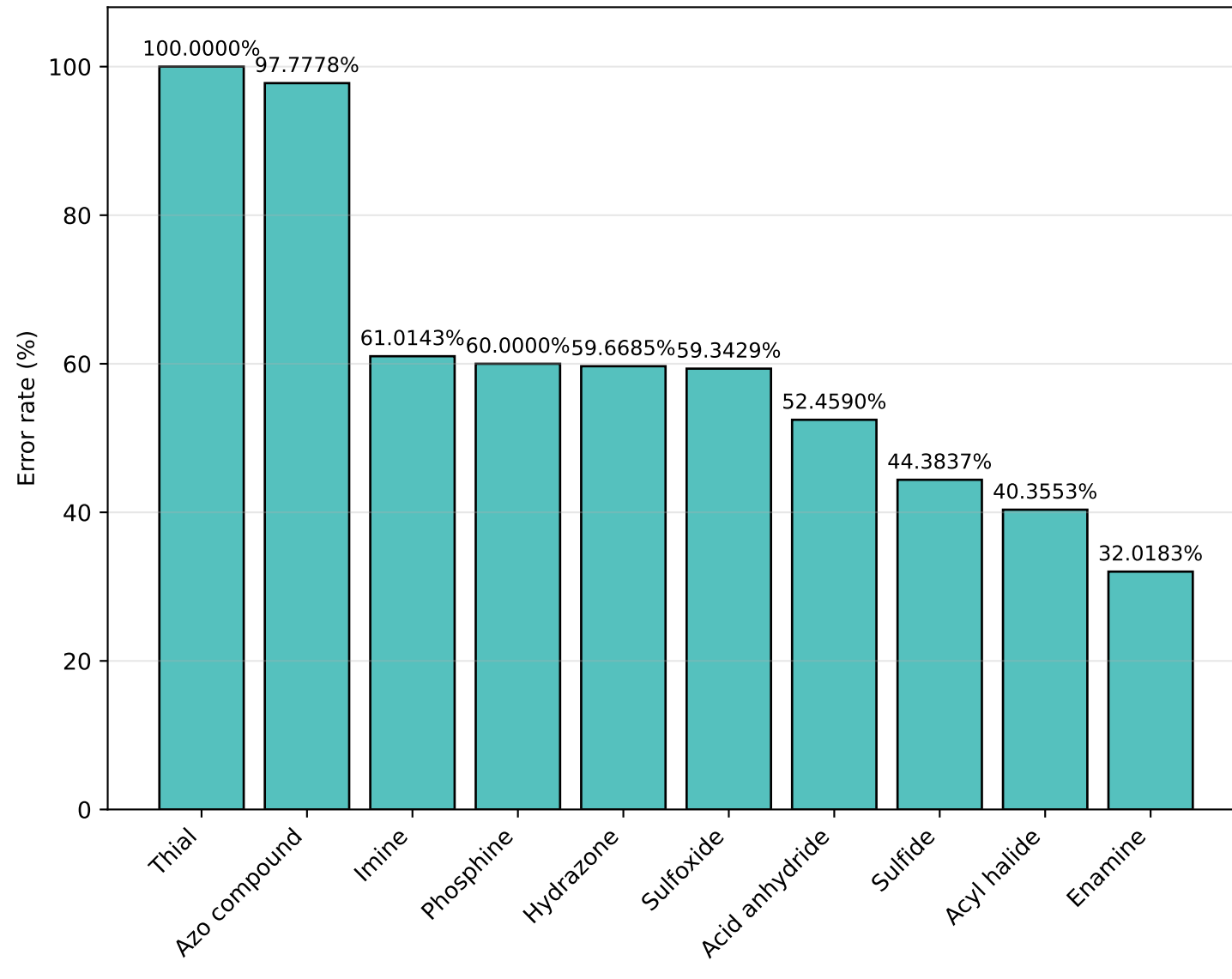
CNN IR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



CNN IR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.9398
F1 macro:	0.6778
Precision micro:	0.9619
Recall micro:	0.9186
Hamming accuracy:	0.9848

Error-rate statistics:

Mean:	24.5181%
Median:	15.5660%
Std:	26.5550%
Max:	100.0000% (Thial)
Min:	0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%):	7 groups
High (20-50%):	10 groups
Moderate (5-20%):	8 groups
Low (<=5%):	12 groups